ME2605 Training Neural Networks: Theory and Practice **@ SLAI**

co-list: DDA6204 Selected Topics-Training Neural Networks: Theory and Practice @ CUHK-SZ

Lecture 1.1 Overview

Ruoyu Sun

Instructor

Ruoyu Sun

孙若愚

sunruoyu@cuhk.edu.cn

Associate Professor
SLAI & CUHK-SZ

Director, AI and Large Foundation Model Center
SRIBD

PREVIOUSLY

UIUC
Assistant Professor

Facebook AI Research (FAIR)
Visiting scientist, full-time

Stanford · MS&E
Postdoc

University of Minnesota · ECE
PhD

Peking University · Mathematics
B.S.

调查

- **Deep-Learning Knowledge**
- Do you know which optimizer was used to train ChatGPT?
- Do you know what normalization layers large-language models use?
- Do you know which initialization NN training uses?

Optimization Knowledge

- Do you understand the meaning of 1/L learning-rate schedule?
- Do you know what a local minimum is?

Hands-on Practice

- • Have you implemented MNIST classification with PyTorch?
- • Have you ever run a Transformer training loop in code?

History of the Course

- First developed in UIUC, in 2017
- Taught a few times at CUHK-SZ
- Previously, mostly focus on theory
- Due to the available hardware, and the simplification of coding, we will add a significant portion of computation

Part I Motivation

Huge Improvement of AI

AI has made tremendous improvement:

Writing code; generating 3D scenes; solve math problems; …

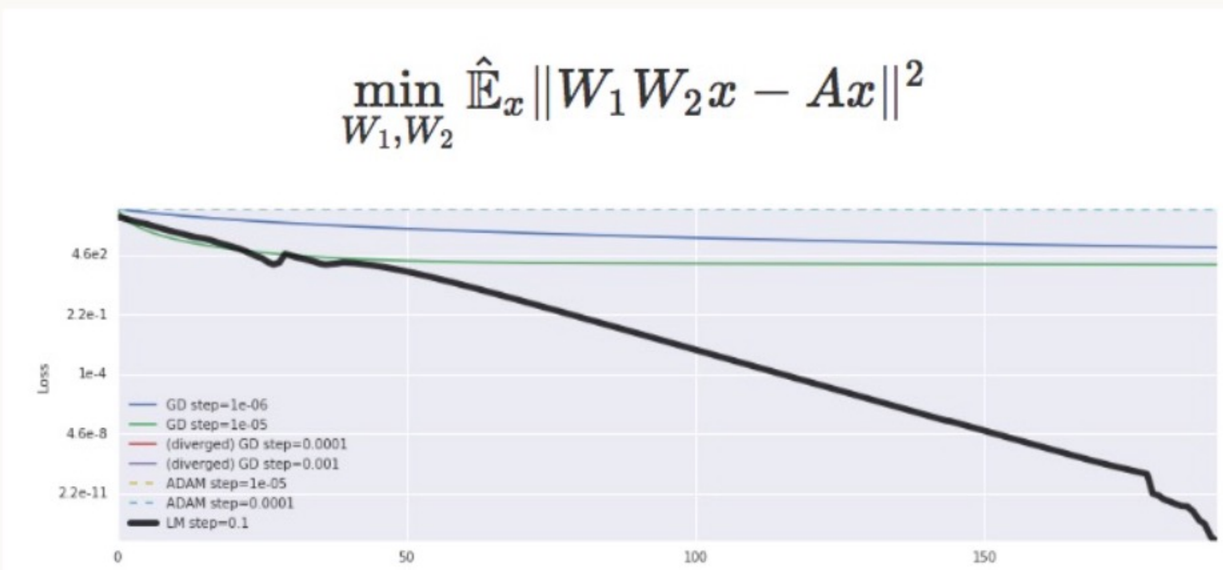
Applications of Neural-nets

Motivation 1: Is AI Alchemy?

NIPS 2017 Test-of-time award presentation,
Rahimi gave a contraversial talk:

Is AI like Alchemy?

Ali Rahimi · NIPS 2017 Test-of-Time Award talk



x: input
A: target linear map
W_1, W_2: trainable matrices
Ê: sample average
‖·‖: Euclidean norm

http://www.argmin.net/2017/12/05/kitchen-sinks/

More Details About the Figure

10^{20}

Condition number of A

The ratio of its largest to smallest singular value.

$\approx 4 \times 10^2$

Loss at the apparent plateau

GD drops quickly at first, then improves very slowly.

Local minimum?	Not established: the gradients are not decaying to zero.
Statistical noise floor?	No: directly minimizing the expected loss shows the same behavior.

Gap

2 layers

Even a linear network can be difficult for GD to train.

1,000 layers

Yet much deeper networks can work well in practice.

How much of this success do we understand? How much comes from tuning?

LeCun’s Comment

“It’s wrong!”

“Ali complained about the lack of (theoretical) understanding in DL.”

“But another important goal is inventing new methods, new tricks, etc.”

“Dangerous (thought). Neural nets, with their non-convex loss functions, had no guarantees of convergence”. “So people discarded it a decade ago.”

https://www.facebook.com/yann.lecun/posts/10154938130592143

Rahimi’s Response and LeCun’s Response

Rahimi: We need simple experiments and simple theorems.

--so that we can be as productive as you guys

LeCun: Possible that no “simple” theorems for neural networks, just as we don’t have analytic solutions to Navier-Stokes equation or the 3-body problem.

Summary of Debate

Do we need theory in AI age?

Rahimi: Current AI research is too empirical;

+: need to understand.

LeCun: No need to have that much (math) theory.

-: NN already works! Why bother "understand"?

Motivation 1 is mainly theoretical….

What applications do neural nets have?

Application of NN

--IT: recommendation; ads

--Transportation: self-driving car

--Health and medical: protein prediction

--Science (physics, biology, etc): drug discovery

--Manufacturing: anomaly detection

Motivation 2: NN for new area

One day, a Physics PhD student comes to ask me: I use a complex-variable neural net for quantum mechanics, but my neural net does not work, why?

I told him: you need to do two things:

i) Diagnosis of the failure: what type?

ii) Try a few common tricks, e.g., increasing width, etc.

Both suggestions require solid understanding of deep learning

Do we need theory?

It takes **decades** to arrive at current DL!

Now, if you want to apply DL to a new area like physics, can you use it well?

Yes, if you have 5 years to explore

Or…. Use existing theory/principles to guide the design

Theory helps you apply DL to physics/medicine/agriculture/etc. !

Summary so far

*We discussed two motivations for **NN theory**:*

Understanding: theoretical understanding of how it works;

Design: based on understanding, design better methods for next generation LLMs or models in other areas

For this reason, we focus on (arguably) the most relevant theory for DL training & practice: optimization theory & algorithms

Part II Course Overview

Intro

Course Material: Course slides.

References:

1) Ongoing book: “Lectures on deep learning optimization”, (title TBD)
2) Research papers in recent years.
3) Classical optimization theory
4) Survey of optimization for DL.
   - R. Sun, “Optimization Theory for Deep Learning: An Overview”, 2019 Aug.
   - Goodfellow, Bengio and Courville, “Deep Learning”, MIT Press, 2016.

Course Form

Lecture; Quiz; Lab practice.

--Encourage discussions.

Will have **a poster session** in the final week.

Grading

In-class quiz: 30%
(NO graded homework)

Midterm examination (Part I and core LLM training concepts): 30%

Course project and poster presentation: 40%
— students choose one of the following:
(a) A pre-training or fine-tuning project, generating a new training pipeline that beats existing baselines on Ascend cards or GPUs.
(b) A theoretical study with experiments on neural net training.
(c) Other types of projects related to training, e.g., benchmark, agent, etc.

Quiz Form

- Quiz website:
  http://47.238.4.125:3100

Very easy to use the quiz website.

Exercise Question Lists

3-5 times (once every 3-4 weeks)

Theory problems + coding problems.

No need to submit solutions.

Mainly for help you grasp the course contents and the preparation of midterm exam.

Project Time Line (tentative)

1-page proposal due: Oct 30.
Describe what you want to do.

Project report due: Dec 16.
10-30 pages.

Poster session: Dec 15, Tues 2:30-5:15pm.

Course group

Homework/PPT:
wechat group.

Other course material:
share via wechat group.

群聊：ME2605_NN_training

该二维码7天内(9月13日前)有效，重新进入将更新

Pre-requisite

- **Prerequisites**:
  - Calculus
  - Linear algebra
  - Probability
  - Basics of optimization (gradient descent)
  - Basic deep learning (linear/logistic regression)

- **Related subjects** (helpful, not required)
  - Basic knowledge of computer vision and NLP
  - Following the trend of DL and AI

Pre-Requisite Part 1: Basic Machine Learning (will review some parts briefly)

- https://github.com/scutan90/DeepLearning-500-questions 500 interview questions for deep learning

- 2.3 监督学习有哪些步骤
- 2.5 分类，回归和聚类的区别和联系？

- 2.12 理解局部最优与全局最优 (简单定义)
- 2.13 理解逻辑回归

- 2.15-18 为什么需要损失函数(loss function)？原理？为何非负？常见例子？

- 2.23 机器学习中为什么需要梯度下降？

Pre-Requisite Part II: Gradient Descent Method (will review some parts briefly)

- 2.25 梯度下降法直观理解？
  2.26 梯度下降法算法描述？
  2.28 随机梯度(stochastic gradient)和批量梯度(batch gradient)区别？

- 梯度下降法“收敛”是什么意思？
- 梯度下降法取什么步长可以保证收敛性？ constant/backtracking/diminishing/etc.;

- SGD什么时候收敛？
- 对于凸函数和强凸函数，GD收敛速度分别是多少？

Topics Covered

- Overview and discussion of DL training methods and underlying theory
- Focus: math behind essential components of deep learning

- Neural-net dependent methods
  - Initialization and signal propagation analysis
  - BatchNorm, ResNet
  - Global optimization: NTK theory and PL condition
- Landscape, non-convex optimization
- Representation and generalization for training neural nets
- SGD variants; Adam, AdaGrad, RMSProp
- LLM basics
- Pretraining a small GPT

Tentative Schedule

Part 0 — Quick guide

W1	Sep 8	Run MLP on MNIST: matrix backprop, cross-entropy, autograd

Part I — Foundations of neural network training

W2	Sep 15	Optimization: GD, SGD + momentum, AdaGrad/RMSProp/Adam
W3	Sep 22	Initialization and signal propagation: Xavier/He, isometry
W4	Sep 29	Normalization (BN/LN/RMSNorm), residual MLPs, landscape
—	Oct 6	National Day break (Oct 1–7) — no class

Part II — Large language model training

W5	Oct 13	Seq2Seq, Transformers, nanoGPT — pre-train a small GPT
W6	Oct 20	SFT and LoRA: instruction formatting, low-rank updates
W7	Oct 27	Distributed training on Ascend 910C: all-reduce, grad accum

Tentative Schedule

Part III — Advanced LLM training

W8	Nov 3	Advanced architectures: sparse attention, GQA, MLA, DSA
W9	Nov 10	Second-order methods: Gauss-Newton, Shampoo, Muon
W10	Nov 17	Hessian structure and spectrum: Hv-products, spectral analysis
W11	Nov 24	Kernels and hardware: FlashAttention, performance tuning
W12	Dec 1	Scaling laws: Kaplan/Chinchilla, model-data-systems tradeoff
W13	Dec 8	Precision and data selection: bf16 vs fp32, error analysis
W14	**Dec 15**	**Final demo — open poster symposium, 3 hours**

课程目标：学到什么？

Main goals:

i) Train an LLM from scratch;

ii) Understand the principles behind major design choices.

- **Intended Audience:**
  - Theorists (optimizers, machine learners, statisticians, mathematicians, theoretical computer scientists, etc.) who want to work on or interested in theory of DL
  - Practitioners keen in theory
  - Domain reserchers who want to aply ML to, e.g., finance, physics, protein prediction, etc.

Will AI Take Over? Decision v.s. Execution

AI can write code. **Will AI take over the training itself**?

Old Story: decision v.s. execution.

$10 vs $990

THE PRICE OF KNOWING WHERE

- **机器故障，无人能修**
  Machine breaks, no one can fix
- **老工程师画一个圈**
  Old engineer draws a circle
- **说：**
  Says:

> **“拧这里”**
> “Tighten here”

📋 **账单** INVOICE

拧螺丝 Turn screw	$10
知道在哪拧 Know where to turn	$990
合计 TOTAL	**$1000**

AI v.s. Human in Neural Net Training

Phase I (now)

AI writes training code

AI 能写训练代码

人类决定改进方向

Humans decide which directions to improve

Phase II (emerging)

AI knows training theory

AI 懂训练理论

人类决定探索什么

Humans decide what to explore

谁买单?
Who pays the bill?

换新优化器?
New optimizer?

改架构?
Modify architecture?

Boss needs a human to decide
老板需要有人拍板

训练神经网络，比纯数学对 AI 更难

Training neural nets is, in some sense, harder for AI than pure math

Path Dependence: The Width of a Horse

Roman chariot 罗马战车 → **Horse carriage** 英国马车沿用 → **Railway gauge** 标准轨距 1435mm → **Shuttle SRB** 航天飞机助推器

Chariot width = the width of two horses → carriages kept it → **standard gauge 1435 mm** → SRB diameter limited by rail tunnels.

Two millennia ago, a horse’s behind capped the most advanced rocket.

两千年前的马屁股，限制了最尖端的火箭。

This is path dependence 路径依赖: a historical accident locks in the design space.
历史上的偶然选择，锁死了后来的设计空间。

The SGD Dividend (1985–2012)

- **A network that SGD could not train did not survive.**

这期间，SGD 跑不出效果的网络就留不下来。

- Had the Transformer been invented 10 years earlier, it would likely have been ignored —

because no one could train it.

假如 Transformer 提前 10 年发明，大概率没人关注——因为训不出来。

Surviving architectures = the ones SGD could train.
留下来的架构 = SGD 训得动的架构。

The Adam Dividend (2012–2022)

- **Most researchers defaulted to Adam / SGD.**

这期间大部分研究者默认用 Adam / SGD。

- If an architecture did not work with Adam, it could not survive.

如果某个架构用 Adam 不 work，它就没法留下来。

The default optimizer quietly decides which ideas survive.
默认优化器悄悄决定了哪些想法能活下来。

The GPU Dependence (2010–2026)

- **A framework that GPUs could not run at scale would not succeed.**

  这期间，GPU 没法大规模跑的框架就不会成功。

- **Speculation**: Hinton’s Capsule Network may never have gotten its chance.

  猜测：Hinton 的 Capsule Network 可能因此从未得到证明自己的机会。

Theory sharpens these claims — that is the purpose of this course.
理论加深我们对这些说法的理解。这正是本课程的目的。

Warning: Theory is Hard

No guarantee that the theory can really help;

Even worse, some theory could be misleading! (SVM v.s. non-convex)

Know partial theory can be WORSE than knowing no theory!
尽信书不如无书。

Respect the complexity of reality, but do not fear it.
尊重现实的复杂性，但不畏惧。

Difference with Most Theoretical Courses

Most math/theory course:

established theory or algorithms.

This course:

--More practical

--Less systematic & ”clean theorems”

--Still a rapidly growing area

Intended Outcome

A well-trained student in the course will:

i) Help train better LLMs in kimi, deepseek, bytedance, Huawei, etc.

ii) Train better embodied AI models, better AI4S models, etc.

iii) Teach future generation of AI students.

Summary

This course is an "advanced topics"-type course.

This course teaches:

Train an LLM from scratch, and know why.

This course does not have tons of theorems, but rather a mix of theory and intuition & practice

---**theory that openAI/deepseek/HW/tencent ppl care**